



Indian Institute of Technology
Madras

Data Science and AI Labs project

AI-Based Forest Cover Change Detection and Monitoring System

Prepared by:

Harsh kumar
Suranjan Das
Aditya Nupani
Subham Mishra
Aman Kankriya

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0.1 Introduction

0.1.1 Background and Motivation

The North Eastern Region (NER) of India is a critical biodiversity hotspot characterized by dense tropical forests that provide essential ecosystem services such as carbon sequestration and habitat preservation. However, these forests are increasingly threatened by urban expansion, infrastructure development, and shifting cultivation practices (Jhum). Existing global datasets, such as the Hansen Global Forest Change dataset, operate at 30-meter resolution and often fail to capture small-scale disturbances common in this region. Additionally, optical imagery is severely limited by persistent cloud cover during the monsoon season, leading to significant data gaps. These limitations motivate the need for a high-resolution, cloud-robust monitoring system tailored to the NER.

0.1.2 Problem Statement

This study addresses the problem of pixel-level forest loss detection using multi-temporal satellite imagery. Given co-registered image pairs from 2021 (T1) and 2023 (T2), the goal is to generate a binary mask identifying forest loss. This is formulated as a supervised binary semantic segmentation task, where Class 0 represents no change and Class 1 represents forest loss. The task is challenging due to severe class imbalance and the presence of both large-scale and small, irregular deforestation patterns.

0.1.3 Challenges in Forest Change Detection

Forest change detection in the NER presents several challenges. Persistent cloud cover limits optical data availability, necessitating the use of SAR imagery. Deforestation occurs at multiple spatial scales, requiring multi-scale feature learning. The dataset is highly imbalanced, with change pixels forming a small minority. Additionally, label noise arises from the mismatch between 30-meter labels and 10-meter input imagery. Temporal variations such as seasonal and atmospheric changes further increase the risk of false detections.

0.1.4 Objectives of the Study

The objectives of this study are fourfold. First, to construct a multi-modal dataset combining Sentinel-1 SAR, Sentinel-2 optical data, and Hansen-derived labels for Meghalaya and Nagaland. Second, to establish a baseline using a Siamese U-Net with ASPP. Third, to explore improvements such as attention mechanisms, feature fusion, and deep supervision through controlled experiments. Finally, to achieve significant performance gains over baseline methods, targeting improved F1-score and IoU for accurate forest loss detection.

0.2 Dataset and Preprocessing

0.2.1 Dataset Description

The dataset used in this study is a multi-modal, multi-temporal satellite dataset constructed for forest change detection in the North Eastern Region of India. Each sample consists of a pair of co-registered image patches acquired at two time steps (2021 and 2023), along with a corresponding binary label mask indicating forest loss. The dataset integrates both radar and optical modalities to ensure robustness under varying environmental conditions such as cloud cover. Each sample is represented as a fixed-size patch of 256×256 pixels, with multiple spectral channels capturing complementary information about vegetation and surface structure.

0.2.2 Study Regions (North-East India)

The study focuses on the states of Meghalaya and Nagaland, which are characterized by dense tropical forests and significant anthropogenic pressure. These regions were selected due to their high prevalence of small-scale deforestation activities such as shifting cultivation (Jhum), which are often missed by coarse-resolution global datasets. The complex terrain, frequent cloud cover, and fragmented forest patterns make these regions particularly suitable for evaluating high-resolution change detection models.

0.2.3 Data Sources

Sentinel-1 (SAR Data)

Sentinel-1 provides Synthetic Aperture Radar (SAR) imagery, which is capable of penetrating cloud cover and operating independently of daylight conditions. This makes it particularly valuable in regions like North East India, where persistent cloud cover limits the usability of optical imagery. SAR data captures structural information of the surface, including vegetation density and moisture content, which is useful for detecting changes in forest cover.

Sentinel-2 (Optical Data)

Sentinel-2 provides high-resolution optical imagery across multiple spectral bands. These bands capture reflectance properties of vegetation, enabling the extraction of features related to plant health, biomass, and land cover type. Optical data complements SAR by providing rich spectral information, although it is sensitive to atmospheric conditions such as clouds and haze.

Hansen Forest Loss Labels

Ground truth labels are derived from the Hansen Global Forest Change dataset, which provides annual forest loss information at a spatial resolution of 30 meters. These labels are processed to generate binary masks corresponding to forest loss between 2021 and 2023. Due to the resolution mismatch between labels (30m) and input imagery (10m), the labels may contain boundary uncertainty and noise.

0.2.4 Data Structure and Format

Each data sample consists of three components:

- $T1$: Multi-channel image patch from year 2021
- $T2$: Multi-channel image patch from year 2023
- Label: Binary mask indicating forest loss

Both $T1$ and $T2$ are represented as tensors of shape (C, H, W) , where C denotes the number of channels (combining SAR and optical bands), and $H = W = 256$. The label is a single-channel binary mask of shape $(1, 256, 256)$.

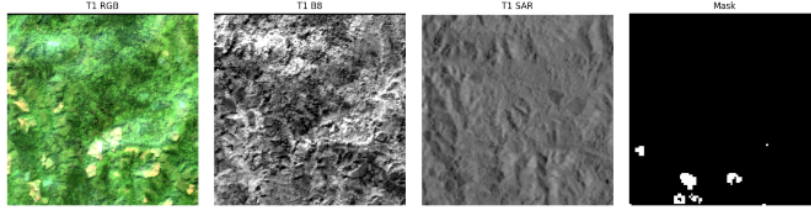


Figure 1: Band visualization

0.2.5 Patch Extraction Strategy

To facilitate efficient training and increase the number of training samples, large satellite images are divided into smaller patches of size 256×256 . This patch-based approach allows the model to focus on localized patterns of change while maintaining manageable computational requirements. It also increases dataset diversity and improves generalization.

0.2.6 Data Normalization

All input channels are normalized using dataset-specific mean and standard deviation values. This standardization ensures that each channel has a comparable scale, which stabilizes training and accelerates convergence. Normalization is particularly important in multi-modal settings where different data sources (SAR and optical) have varying value ranges.

0.2.7 Class Imbalance Analysis

Forest change detection datasets are inherently imbalanced, with the majority of pixels belonging to the no-change class. In the constructed dataset, change pixels represent a small fraction of the total pixel distribution. This imbalance can bias the model toward predicting the majority class, leading to poor detection performance on minority (change) regions. Addressing this imbalance is a key consideration in model design and loss function selection.

0.2.8 Change Ratio Distribution and Binning

To better understand the distribution of change across samples, the ratio of change pixels to total pixels is computed for each patch. Based on this ratio, samples

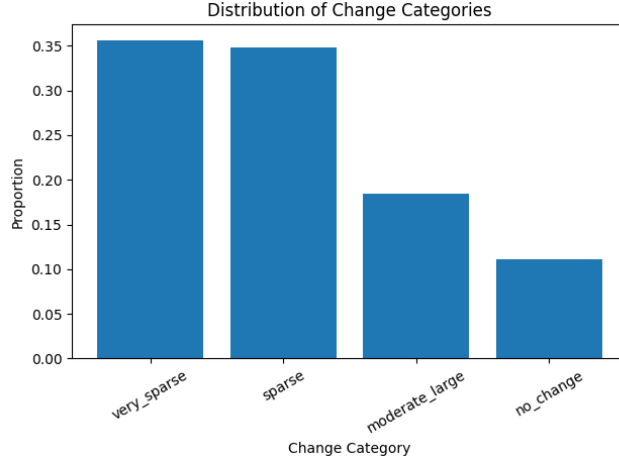


Figure 2: Distribution of samples across change intensity bins showing class imbalance in the dataset.

are categorized into four bins: no change, very sparse change, sparse change, and moderate-to-large change. This binning strategy enables stratified sampling and ensures that the dataset captures a balanced representation of different change intensities.

0.2.9 Train-Validation-Test Split Strategy

The dataset is split into training, validation, and test sets using a stratified sampling approach based on region and change intensity bins. This ensures that each split maintains a similar distribution of change patterns, preventing bias toward specific regions or change types. A typical split ratio of 70% training, 15% validation, and 15% testing is used to balance model training and evaluation.

0.3 Baseline Model Architecture

0.3.1 Overview of U-Net

U-Net is an encoder-decoder architecture originally proposed for biomedical image segmentation. It consists of a contracting path that captures contextual features through successive convolutions and pooling, and an expansive path that recovers spatial resolution through upsampling. Skip connections bridge corresponding encoder and decoder levels, preserving fine-grained spatial detail that would otherwise be lost during downsampling.

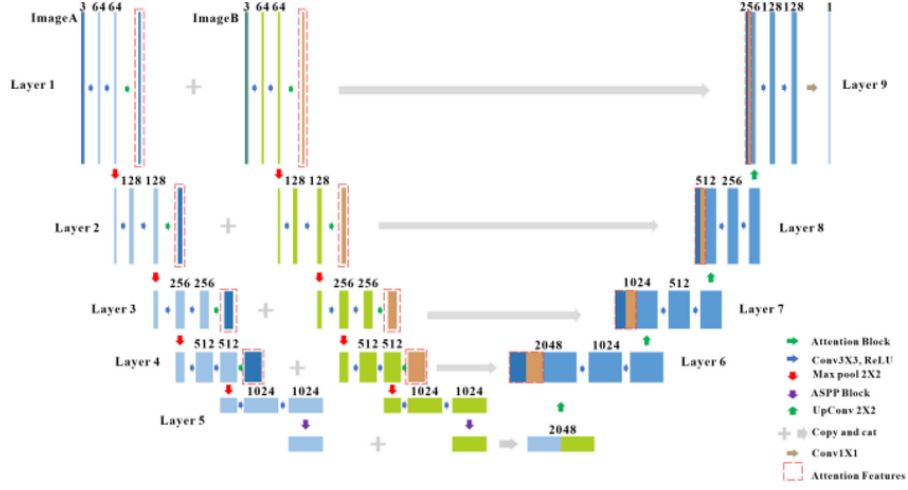


Figure 3: Structure of baseline model

0.3.2 Siamese Network Concept

A Siamese network consists of two branches sharing identical weights and architecture, designed to process two inputs simultaneously and compare their representations. In the context of change detection, each branch processes one temporal image independently, and the resulting feature maps are compared to identify regions of change. Weight sharing ensures that both branches extract features in a consistent representation space, making temporal differences semantically meaningful.

0.3.3 Siamese U-Net Architecture

The baseline model combines the Siamese concept with a U-Net backbone. Both temporal images — T1 (2021) and T2 (2023) — are passed through the same shared encoder, producing multi-scale feature maps at four resolution levels. These features are fused and decoded to produce a binary change mask at the original input resolution of 256×256 pixels.

0.3.4 Encoder Design

The encoder consists of four DoubleConv blocks, each comprising two consecutive Conv2d-BatchNorm-ReLU operations, followed by MaxPool2d downsampling. The channel progression follows $6 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$, where 6 corresponds to the six input channels (S1 VV, VH and S2 B2, B3, B4, B8).

0.3.5 Decoder Design

The decoder consists of four **UpBlock** modules, each applying a transposed convolution for upsampling followed by a **DoubleConv** block. The channel progression is reversed from $512 \rightarrow 256 \rightarrow 128 \rightarrow 64$, culminating in a 1×1 convolution that produces the final single-channel binary prediction.

0.3.6 Skip Connections

At each decoder level, the upsampled feature map is concatenated with the corresponding skip features from both T1 and T2 encoder branches, providing the decoder with both high-level semantic context and low-level spatial detail from both temporal images simultaneously.

0.3.7 Feature Difference Strategy

Rather than subtracting two independently produced segmentation maps, the model concatenates T1 and T2 encoder features directly at each decoder level. This allows the network to learn change representations end-to-end at the feature level, avoiding the compounding classification errors inherent in post-classification comparison approaches.

0.3.8 ASPP (Atrous Spatial Pyramid Pooling)

Motivation for ASPP

Forest loss events in the NER vary significantly in spatial extent, from large contiguous clearings to narrow Jhum patches spanning only a few pixels. A standard bottleneck convolution operates at a single fixed receptive field and cannot simultaneously capture both fine-grained boundaries and broad contextual patterns.

Multi-scale Feature Extraction

ASPP addresses this by applying four parallel atrous convolutions with dilation rates of 1, 6, 12, and 18 to the bottleneck feature map, capturing context at progressively larger receptive fields without reducing spatial resolution. The outputs are concatenated and projected via a 1×1 convolution to a unified 512-channel representation.

Comparison with Standard CNN

Unlike a standard CNN bottleneck which uses a single receptive field, ASPP aggregates multi-scale context in a single forward pass with no additional parameters for downsampling, preserving spatial resolution at the deepest encoder level and significantly improving detection of both small-scale Jhum patches and large-scale deforestation events.

Stage	Layer	Input Channels	Output Channels
Encoder			
Conv Block 1	DoubleConv	6	64
Conv Block 2	DoubleConv	64	128
Conv Block 3	DoubleConv	128	256
Conv Block 4	DoubleConv	256	512
Pooling	MaxPool	–	–
ASPP Module			
1×1 Conv	Conv2d	512	512
3×3 Conv (d=6)	Conv2d	512	512
3×3 Conv (d=12)	Conv2d	512	512
3×3 Conv (d=18)	Conv2d	512	512
Projection	Conv2d	2048	512
Decoder			
UpBlock 4	TransposeConv + DoubleConv	1536	512
UpBlock 3	TransposeConv + DoubleConv	768	256
UpBlock 2	TransposeConv + DoubleConv	384	128
UpBlock 1	TransposeConv + DoubleConv	192	64
Output			
Final Layer	Conv2d (1×1)	64	1

Table 1: Architecture of Siamese U-Net with ASPP

0.4 Experiments

0.4.1 Loss Functions

Binary Cross Entropy

Binary Cross Entropy (BCE) was used as the baseline loss function. While effective for balanced binary classification, BCE treats all pixels equally and is highly sensitive to class imbalance. In our dataset where forest loss pixels constitute a small

minority, BCE-only training causes the model to favour the background class, producing near-zero recall for the change class.

Dice Loss

Dice Loss directly optimizes the overlap between predicted and ground truth masks, making it naturally robust to class imbalance. It is defined as:

$$\mathcal{L}_{Dice} = 1 - \frac{2 \cdot |P \cap T|}{|P| + |T|} \quad (1)$$

where P is the predicted mask and T is the ground truth mask. By focusing on the ratio of overlap rather than pixel-wise agreement, Dice Loss penalizes missed detections more strongly than BCE, which is particularly beneficial for sparse forest loss signals.

Focal Loss

Focal Loss addresses class imbalance by down-weighting the loss contribution of easy-to-classify background pixels and focusing training on hard positives. It modifies BCE as:

$$\mathcal{L}_{Focal} = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (2)$$

where $\alpha = 0.8$ controls the class weighting and $\gamma = 2$ is the focusing parameter. Pixels the model classifies confidently contribute less to the total loss, directing gradient updates toward the ambiguous forest loss boundaries.

Combined Loss and Weighting Strategy

Two loss combinations were evaluated across experiments. The first combined BCE and Dice equally ($0.5 \cdot \mathcal{L}_{Dice} + 0.5 \cdot \mathcal{L}_{BCE}$), used in the initial baseline run. The second replaced BCE with Focal Loss and reweighted to ($0.7 \cdot \mathcal{L}_{Dice} + 0.3 \cdot \mathcal{L}_{Focal}$), giving Dice Loss dominance to preserve recall while Focal Loss suppresses background noise. The Dice+Focal combination yielded measurable improvements in both F1 and IoU over the BCE+Dice baseline and was adopted as the standard loss function for all subsequent experiments.

0.4.2 Training Strategy and Experimental Setup

Pipeline, Optimizer and Mixed Precision

All models were trained on a Kaggle notebook with an NVIDIA Tesla T4 GPU (15GB VRAM) using PyTorch. Training employed Automatic Mixed Precision (AMP) via `torch.amp.autocast` and `GradScaler`, which reduces memory consumption and accelerates computation by performing forward passes in float16 while maintaining float32 precision for gradient accumulation. The Adam optimizer was used with an initial learning rate of 1×10^{-4} and default momentum parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$). A `ReduceLROnPlateau` scheduler monitored validation IoU with a reduction factor of 0.5 and patience of 2 epochs, halving the learning rate whenever no improvement was observed. Early stopping with a patience of 5 epochs prevented overfitting and reduced unnecessary compute. All experiments used a batch size of 16 and were allowed a maximum of 50 epochs, with the best checkpoint selected by highest validation IoU.

Hyperparameter Configuration

Hyperparameter	Value
Optimizer	Adam
Learning rate	1×10^{-4}
Batch size	16
Max epochs	50
LR scheduler	ReduceLROnPlateau
LR reduction factor	0.5
Scheduler patience	2 epochs
Early stopping patience	5 epochs
Mixed precision	AMP (float16/float32)
Input patch size	256×256
Input channels	6 (S1 VV, VH + S2 B2, B3, B4, B8)

Table 2: Fixed hyperparameter configuration across all experiments

Experiment Tracking and Comparison Strategy

All experiments were tracked using Weights & Biases (W&B), logging training loss, validation F1, validation IoU, and learning rate at every epoch. Best model checkpoints were saved as W&B artifacts for reproducibility. Each experiment was run independently with identical hyperparameters and data splits, changing only the model architecture or loss function, ensuring that performance differences are attributable solely to the modification under study. Results are compared across runs using validation F1 and IoU as primary metrics, with the held-out test set reserved for final evaluation only.

0.4.3 Evaluation Metrics

All metrics are computed at the pixel level on binary predictions obtained by thresholding sigmoid-activated logits. Let TP, FP, FN, and TN denote true positives, false positives, false negatives, and true negatives respectively.

Precision, Recall and F1 Score

Precision measures the fraction of predicted change pixels that are genuinely changed, penalizing false alarms: $\text{Precision} = \frac{TP}{TP+FP}$. Recall measures the fraction of actual change pixels correctly detected, penalizing missed detections: $\text{Recall} = \frac{TP}{TP+FN}$. The F1 Score is their harmonic mean and serves as the primary evaluation metric given the class imbalance:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

Intersection over Union (IoU)

IoU measures the spatial overlap between predicted and ground truth change regions and is used as the secondary metric and the criterion for model checkpointing:

$$IoU = \frac{TP}{TP + FP + FN} \quad (4)$$

IoU is stricter than F1 as it penalizes both false positives and false negatives simultaneously without the compensating effect of true negatives.

Threshold Selection

Rather than using a fixed threshold of 0.5, an optimal decision threshold is searched over the range $[0.10, 0.55]$ in steps of 0.05 on the validation set predictions at the end of training, selecting the value that maximizes F1. This accounts for the class imbalance which typically shifts the optimal threshold below 0.5, and ensures that reported metrics reflect the best achievable performance for each model configuration.

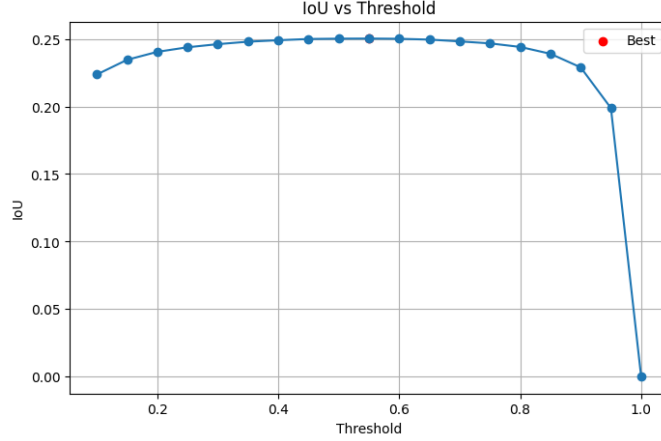


Figure 4: Variation of IoU over threshold

0.5 Model Enhancement Techniques

0.5.1 Training-Level Improvements

The first set of enhancements targeted the training pipeline rather than the model architecture. Random horizontal and vertical flips along with 90 rotations were introduced as data augmentation strategies, increasing the diversity of training samples and improving generalization to the varied spatial orientations in which deforestation patterns appear across Meghalaya and Nagaland. The loss function was refined by replacing the BCE+Dice combination with a weighted Dice+Focal formulation ($0.7 \cdot \mathcal{L}_{Dice} + 0.3 \cdot \mathcal{L}_{Focal}$), better handling class imbalance by directing gradient updates toward hard-to-detect forest loss pixels. A dynamic threshold optimization strategy was also introduced, searching for the decision threshold that maximizes F1-score on the validation set rather than using a fixed value of 0.5. Finally, a `ReduceLROnPlateau` scheduler was employed to adaptively halve the

learning rate upon validation stagnation, enabling more stable convergence in later epochs. Together these changes improved F1 from 0.4930 to 0.5132 and IoU from 0.2921 to 0.3475.

0.5.2 Pretrained Encoder: PretrainedSiameseUNet_ASPP

The second experiment replaced the from-scratch VGG-style encoder with pre-trained ResNet-34 layers while keeping the ASPP bottleneck, all four UpBlocks, and the final head completely intact. ResNet-34 was selected because its stage output channels — 64, 64, 128, 256, and 512 — match the original encoder channel progression exactly, requiring zero modifications downstream.

The encoder is organized into five explicit stages: **enc0** (conv1 + BN + ReLU, 64ch at 128×128), **enc1** (maxpool + layer1, 64ch at 64×64), **enc2** (layer2, 128ch at 32×32), **enc3** (layer3, 256ch at 16×16), and **enc4** (layer4, 512ch at 8×8). Both temporal branches share the same encoder object, preserving the Siamese weight-sharing property.

The primary challenge was the input channel mismatch between ImageNet-pretrained weights (3-channel RGB) and our 6-channel satellite input (S1 VV/VH + S2 B2/B3/B4/B8). This was resolved by replacing the first convolution with a new Conv2d(6, 64, 7×7) whose weights were initialized by averaging the pre-trained 3-channel kernel across the input dimension and tiling it across all 6 channels. This preserves pretrained spatial filters in all deeper layers while adapting only the input layer to the new modality. The core motivation is that training from scratch on 7,500 samples provides insufficient signal for the encoder to learn robust low-level feature representations, whereas a pretrained ResNet-34 starts with weights already sensitive to edges, textures, and gradients that transfer effectively to satellite imagery.

0.5.3 Architectural Improvements: SiameseUNet_V2

The third experiment introduced three structural improvements into the decoder path of the original model while leaving the encoder and ASPP unchanged.

Temporal Bottleneck Fusion. Before entering the ASPP, features from both temporal branches are concatenated to 1024 channels and projected back to 512ch via a 1×1 Conv-BN-ReLU block, making the bottleneck temporally aware prior

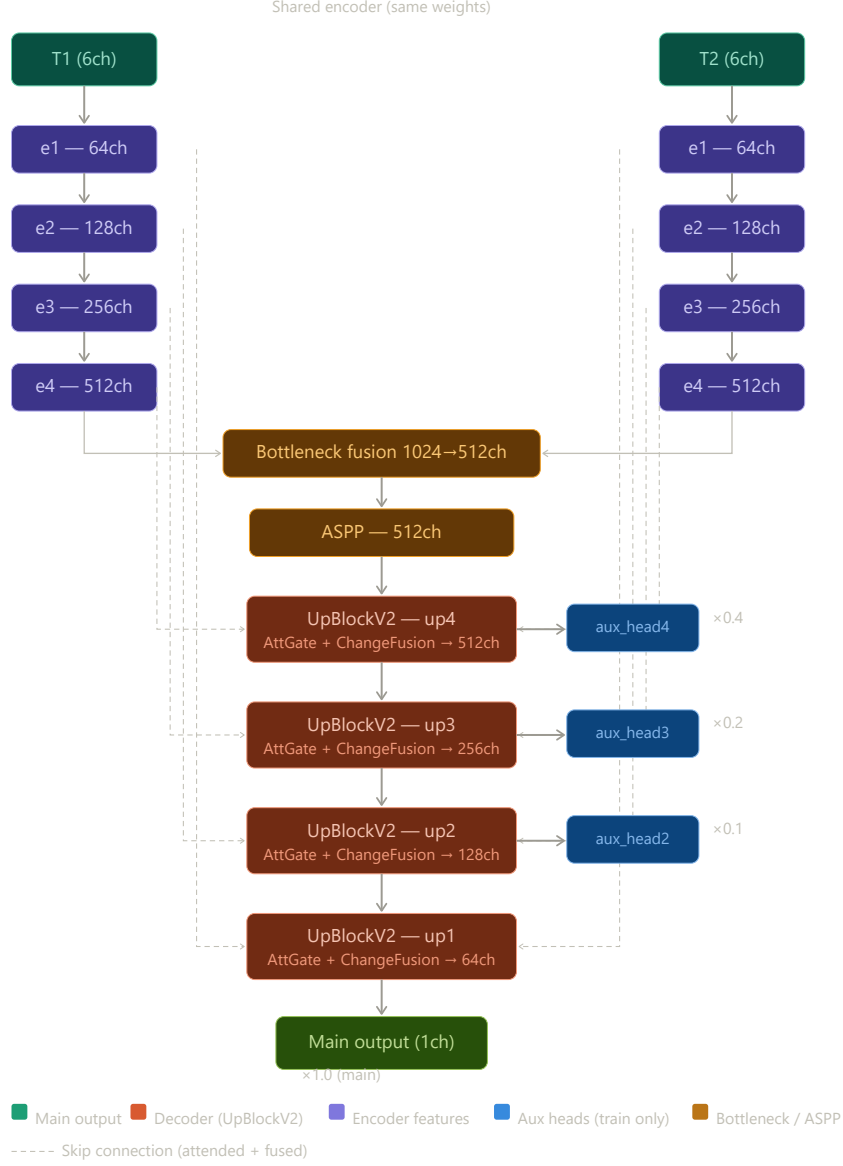


Figure 5: Structure of SiameseUNet_V2 model

to multiscale feature extraction.

Attention Gates and Change Feature Fusion. Each skip connection is passed through a dedicated Attention Gate that uses the upsampled decoder feature as a gating signal, producing a soft spatial mask that suppresses background activations — critical given that approximately 89% of pixels are unchanged. The attended features are then passed through a **ChangeFusion** module that explicitly computes $[t_1, t_2, |t_1 - t_2|, t_1 \cdot t_2]$, concatenates them to $4 \times \text{ch}$, and projects back to $2 \times \text{ch}$. This provides the decoder with an explicit change signal encoding both the magnitude of difference and feature correlation, directly addressing the weak contrast produced by small Jhum patches and sparse deforestation events.

Deep Supervision. Three auxiliary 1×1 convolutional heads are attached to intermediate decoder outputs **d4**, **d3**, and **d2**. Each produces a full-resolution prediction contributing to the total loss with weights 0.4, 0.2, and 0.1 respectively alongside the main head at weight 1.0. This injects gradient signal directly into early decoder layers, addressing the training plateau observed at epochs 25–30 in prior runs. All auxiliary heads are bypassed at inference via a `self.training` guard, adding zero computational overhead during evaluation.

0.6 Results and Analysis

0.6.1 Baseline Model Performance

The baseline `SiameseUNet_ASPP` model trained from scratch using a combined BCE and Dice loss ($0.5 \cdot \mathcal{L}_{Dice} + 0.5 \cdot \mathcal{L}_{BCE}$) was evaluated over a maximum of 50 epochs with early stopping. Training terminated at epoch 19 after 5 consecutive epochs without improvement in validation IoU, with the best checkpoint recorded at epoch 14.

Table 3: Baseline model training summary

Metric	Value
Best Validation F1	0.5059
Best Validation IoU	0.3413
Final Training Loss	0.3517
Epochs to Best	14
Total Epochs Trained	19
Early Stopping	Yes (patience = 5)

Training Dynamics Analysis

As shown in Figure 8, both validation F1 and IoU exhibit a broadly increasing trend across the first 14 epochs, with notable instability in the early phase. A sharp drop is observed at epoch 3 followed by recovery, and again at epoch 7 where F1 fell from 0.483 to 0.424 and IoU from 0.321 to 0.271 before recovering in the



Figure 6: *

(a) Validation F1 Score



Figure 7: *

(b) Validation IoU

Figure 8: Validation F1 and IoU curves for the baseline `SiameseUNet_ASPP` model across 19 training epochs.

following epoch. This oscillatory behaviour in early training is characteristic of from-scratch initialization on a small, class-imbalanced dataset, where the model alternates between over-predicting the majority background class and recovering toward the minority forest loss class. From epoch 9 onward, both curves stabilize and plateau in a narrow band — F1 between 0.491 and 0.506, IoU between 0.328 and 0.341 — indicating that the model has reached the limit of its representational capacity under the current configuration. The training loss decreases monotonically from 0.753 to 0.352, confirming that the model continues to fit the training data after validation performance saturates, which is consistent with the observed early stopping trigger.

0.6.2 Model Enhancement: Data Augmentation and Loss Refinement

Building on the baseline, the first enhancement experiment introduced random horizontal and vertical flips, 90 rotations, a refined Dice+Focal loss ($0.7 \cdot \mathcal{L}_{Dice} + 0.3 \cdot \mathcal{L}_{Focal}$), dynamic threshold optimization, and a learning rate scheduler. Despite these improvements, the gains over the baseline were marginal, with the best checkpoint recorded at epoch 23.

Metric	Baseline (BCE+Dice)	Enhanced (Dice+Focal)
Best Validation F1	0.5060	0.5189
Best Validation IoU	0.3413	0.3526
Epochs to Best	14	23
Total Epochs Trained	19	30

Table 4: Performance comparison: Baseline vs. Enhancement experiment

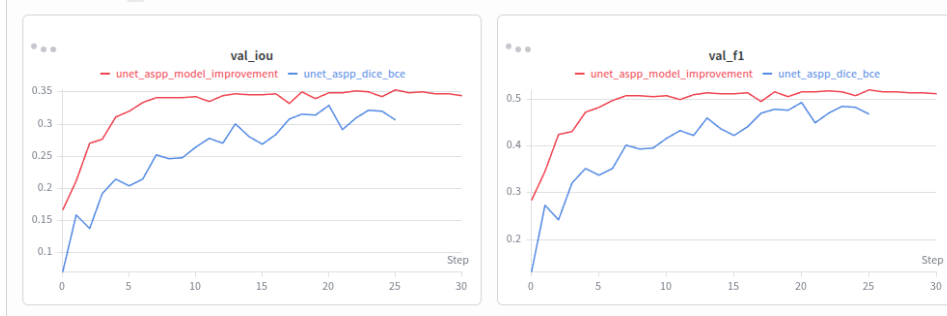


Figure 9: Validation IoU (left) and F1 (right) comparison between the baseline `UNET_ASPP_DICE_BCE` (blue) and the enhanced model `UNET_ASPP_MODEL_IMPROVEMENT` (red).

Analysis

As shown in Figure 9, the enhanced model (red) consistently outperforms the baseline (blue) across all epochs in both F1 and IoU. The most notable difference is in the early training phase — the enhanced model reaches a validation IoU of approximately 0.27 by epoch 1 and F1 of 0.42 by epoch 3, whereas the baseline required several more epochs to achieve comparable values. This faster initial convergence is attributable to the Focal Loss component directing gradient updates toward hard positive forest loss pixels from the start of training, combined with augmentation providing more varied spatial configurations per epoch. However, both models converge to a similar performance ceiling, with the enhanced model improving F1 by only 0.013 and IoU by 0.011 over the baseline. The plateau behaviour persists, now occurring around epoch 23–25 rather than epoch 14, suggesting that the training-level improvements delay saturation but do not fundamentally overcome the representational bottleneck of the from-scratch encoder. This observation directly motivates the architectural experiments explored in subsequent sections.

0.6.3 Pretrained ResNet-34 Encoder

The second architectural experiment replaced the from-scratch VGG-style encoder with a pretrained ResNet-34 backbone while keeping the ASPP and decoder completely unchanged. Despite the stronger initialization from ImageNet weights, the model underperformed relative to the baseline, achieving a best validation F1 of 0.4625 and IoU of 0.3026 at epoch 12, after which early stopping was triggered at epoch 17. The training loss decreased steadily from 0.7115 to 0.2706, yet validation metrics plateaued and began oscillating after epoch 12, indicating a growing train-validation gap. This counterintuitive result is likely attributable to the domain gap between natural ImageNet features and multi-modal satellite imagery comprising SAR and multispectral bands, combined with the approximate 6-channel weight initialization strategy which may have introduced noise into the early convolutional layers that propagated through training.

0.6.4 SiameseUNet_V2 with Attention, Change Fusion and Deep Supervision

The third experiment introduced temporal bottleneck fusion, attention-gated skip connections, explicit change feature fusion, and deep supervision into the decoder path. Despite the substantial architectural additions, the model achieved a best validation F1 of 0.4869 and IoU of 0.3240 at epoch 12, with early stopping triggered at epoch 17. The higher initial training loss (1.08 at epoch 1) reflects the additional complexity of the deep supervision auxiliary losses, and the model showed consistently slower epoch times ($\approx 3\text{m } 40\text{s}$ versus $\approx 1\text{m}$ for the baseline) due to the attention and fusion modules. While the model showed a steady improvement trajectory up to epoch 12, it failed to surpass the enhanced baseline, suggesting that the added modules increase optimization difficulty when trained from scratch on a relatively small dataset of 7,500 samples.

0.6.5 Overall Comparison and Discussion

Table 0.6.5 summarises the performance of all four experiments across consistent evaluation conditions.

As shown in Figure 10 and Table 0.6.5, the enhanced baseline with Dice+Focal



Figure 10: Validation IoU (left) and F1 (right) across all four experiments. Red: enhanced baseline (`unet_aspp_model_improvement`), dark blue: baseline (`unet_aspp_dice_bce`), light blue: pretrained ResNet-34 (`siamese_resnet34_pretrained`), pink: SiameseUNet_V2 (`siamese_v2_attn_changefusion_deepsup`).

Model	Best F1	Best IoU	Best Epoch	Total Epochs
Baseline (BCE+Dice)	0.5060	0.3413	14	19
Enhanced (Dice+Focal + Aug.)	0.5189	0.3526	23	30
Pretrained ResNet-34	0.4625	0.3026	12	17
SiameseUNet_V2 (Attn+CF+DS)	0.4869	0.3240	12	17

Table 5: Overall comparison of all experiments

loss and data augmentation achieved the best overall performance, marginally improving over the original baseline by 0.013 in F1 and 0.011 in IoU. Notably, both the pretrained ResNet-34 and SiameseUNet_V2 experiments produced results below the baseline, which is a significant and instructive finding. Three factors likely contribute to this outcome. First, all models share the same small training dataset of 7,500 samples, which is insufficient to fully leverage the capacity of more complex architectures. Second, the domain gap between ImageNet pretraining and 6-channel SAR+optical satellite imagery limits the transferability of pretrained features. Third, the additional parameters introduced by attention gates, change fusion modules, and auxiliary heads increase the optimization burden, requiring more data or longer training schedules than the current setup permits. Collectively, these results suggest that for this dataset scale, training stability and loss design have a greater impact on performance than architectural complexity, and that future improvements should focus on dataset expansion or semi-supervised

approaches before further architectural scaling.

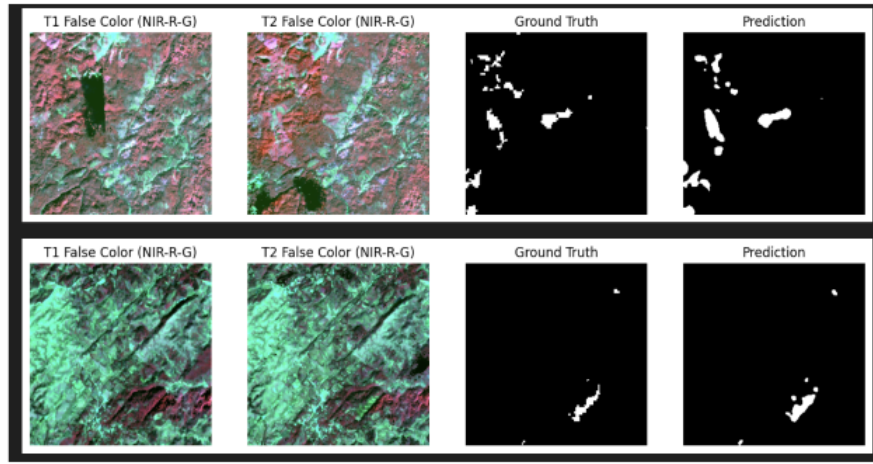


Figure 11: Comparison of the output produced by the baseline model with ground truth

0.7 Conclusion

This study presented a systematic investigation into AI-based forest cover change detection in the North Eastern Region of India, specifically targeting the Meghalaya and Nagaland states over a 2021–2023 temporal window. A multi-modal dataset of 7,500 bi-temporal patches was constructed by fusing Sentinel-1 SAR and Sentinel-2 optical imagery with Hansen Global Forest Change labels, and a baseline *SiameseUNet_ASPP* model was established as the quantitative reference point.

Four experiments were conducted in a controlled, step-by-step manner to isolate the contribution of each modification. The results demonstrate that training-level improvements — specifically the combination of Dice and Focal loss with data augmentation and dynamic threshold optimization — provided the most consistent gains, yielding the best observed F1 of 0.5189 and IoU of 0.3526. Architectural modifications including a pretrained ResNet-34 encoder and the attention-augmented *SiameseUNet_V2* did not surpass the baseline under the current dataset scale, highlighting that model complexity alone is insufficient when training data is limited.

Future work will focus on dataset expansion through additional geographic regions and temporal windows, exploration of semi-supervised or self-supervised pre-training on unlabelled satellite imagery, and integration of purpose-built change detection transformers such as BIT which are designed specifically for bi-temporal

remote sensing tasks. A combination of the most effective strategies from this milestone will serve as the starting point for the final modelling phase.